

UDINE RIVOLTO, Italy

WMO: 160450

Lat: 45.979N

Lon: 13.049E

Elev: 55

StdP: 100.67

Time Zone: 1.00 (EUC)

Period: 90-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.8	-3.2	-14.9	1.0	3.2	-12.2	1.3	1.3	10.2	4.6	8.9	4.5	2.2	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.8	22.8	31.2	22.3	30.0	21.8	24.6	29.8	23.6	28.8	22.8	27.9	2.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.9	27.9	22.0	16.7	26.5	21.1	15.9	25.7	74.6	30.0	70.6	28.8	67.5	27.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.7	5.4	-8.1	35.1	1.8	1.1	-9.4	35.9	-10.5	36.6	-11.5	37.2	-12.9	38.0		
(5)			-8.8	26.3	1.5	1.3	-9.8	27.3	-10.7	28.0	-11.6	28.8	-12.7	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	4.6	5.4	9.5	12.9	18.3	21.3	23.7	23.7	18.4	14.2	9.2	4.8
	DBStd	7.56	2.81	3.35	3.23	2.98	3.20	3.17	2.81	2.63	2.78	3.30	3.75	3.01
	HDD10.0	582	169	131	49	7	0	0	0	0	0	6	59	162
	HDD18.3	2133	426	362	272	165	38	9	1	1	32	133	274	419
	CDD10.0	2004	1	2	35	93	257	338	426	426	253	135	35	2
	CDD18.3	512	0	0	0	2	37	97	168	169	36	4	0	0
	CDH23.3	4528	0	0	1	27	268	783	1607	1585	225	32	0	0
	CDH26.7	1513	0	0	0	1	47	229	604	602	28	1	0	0
Wind	WSAvg	2.1	2.1	2.0	2.5	2.4	2.3	2.0	2.1	2.0	2.0	2.1	2.0	2.2
Precipitation	PrecAvg	1083	64	55	64	103	87	84	84	88	120	120	116	78
	PrecMax	1518	228	242	201	254	200	225	157	211	318	301	352	181
	PrecMin	735	6	3	0	16	13	14	27	26	32	9	8	5
	PrecStd	206	57	70	53	62	49	46	38	58	70	81	74	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	18.4	22.8	26.8	30.8	33.6	34.5	34.9	29.0	26.3	20.3	15.6
		MCWB	10.1	10.6	12.6	16.7	21.3	24.2	23.2	22.9	20.8	18.2	15.1	12.3
	2%	DB	12.1	15.0	20.0	24.7	27.8	30.9	32.5	33.0	27.4	24.0	18.0	12.9
		MCWB	8.9	8.9	12.1	16.0	19.3	22.3	22.8	22.7	19.8	17.0	14.6	9.4
	5%	DB	10.9	13.1	17.6	22.1	26.0	29.3	31.3	31.2	25.9	22.1	16.2	10.9
		MCWB	8.2	8.2	11.1	14.5	18.1	21.6	22.4	22.2	19.1	16.2	13.5	7.8
	10%	DB	9.6	11.2	15.1	19.9	24.7	27.6	30.0	30.0	24.4	20.0	14.9	9.5
		MCWB	7.1	7.5	10.3	13.5	17.6	20.7	22.1	21.7	18.6	15.8	12.3	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.1	14.1	17.6	22.2	24.7	25.9	26.1	23.1	20.0	17.0	13.7
		MCDB	13.4	15.6	19.9	25.0	27.8	30.2	31.8	30.5	26.3	22.7	19.7	15.0
	2%	WB	10.1	10.1	13.0	16.3	21.1	23.6	24.6	24.6	21.7	18.6	15.1	9.9
		MCDB	11.3	14.1	17.4	23.2	26.1	28.7	30.1	29.8	25.5	21.9	17.0	11.1
	5%	WB	8.8	9.0	12.1	15.2	19.8	22.6	23.6	23.6	20.5	17.6	13.9	8.6
		MCDB	10.3	12.0	15.8	21.0	24.1	27.5	28.9	28.7	24.0	20.8	15.8	10.2
	10%	WB	7.4	7.9	11.2	14.1	18.7	21.7	22.7	22.7	19.4	16.2	12.6	7.6
		MCDB	8.9	10.3	14.3	18.5	22.8	26.4	28.0	27.8	22.8	19.1	14.5	9.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.7	10.1	9.9	10.0	9.6	10.6	11.0	10.3	9.6	8.7	8.3
		MCDBR	10.2	13.3	13.6	13.9	11.9	11.5	11.8	12.3	11.6	13.1	10.2	10.4
		MCWBR	7.3	8.4	7.7	7.0	5.4	5.0	4.8	5.4	5.5	7.3	6.8	7.7
	5% WB	MCDBR	7.5	10.9	11.1	12.1	9.5	10.4	10.6	11.4	9.5	9.8	7.6	7.8
		MCWBR	6.2	7.5	7.0	6.9	5.4	5.4	5.2	5.8	5.9	6.5	5.6	6.3
Clear-Sky Solar Irradiance	taub	0.335	0.362	0.410	0.453	0.462	0.478	0.465	0.456	0.435	0.415	0.368	0.331	
	taud	2.406	2.322	2.195	2.102	2.127	2.132	2.177	2.215	2.242	2.259	2.339	2.432	
	Ebn at Noon	756	802	808	805	811	798	803	794	774	726	699	718	
	Edh at Noon	73	96	125	149	150	151	142	132	118	101	78	65	
All-Sky Solar Radiation	RadAvg	1.39	2.25	3.47	4.49	5.43	6.14	6.24	5.38	3.95	2.51	1.44	1.21	
	RadStd	0.20	0.36	0.44	0.54	0.58	0.39	0.37	0.46	0.37	0.28	0.23	0.16	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.45	N/A	N/A	+0.68	+0.40	N/A	-123	+94	+56

Nomenclature: See separate page